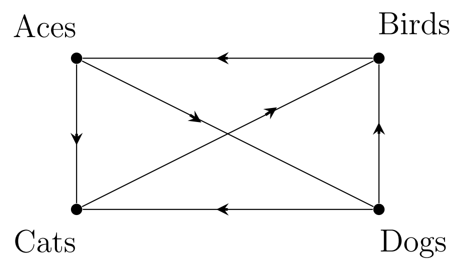
Zack Abdirashid
1/20/26
MATH 405

1.1

2. (a) *Suppose four teams, the Aces, the Birds, the Cats, and the Dogs, play each other once. The Aces beat all three opponents except the Birds. The Birds lose to all opponents except the Aces. The Dogs beat the Cats. Represent the results of these games with a directed graph.*

* A vertex x pointing to a vertex y implies x beat y.

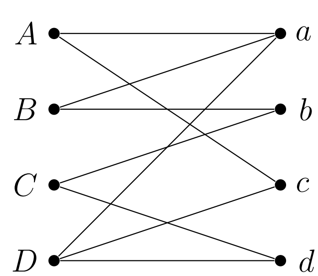


(b) *A dominance order is a listing of teams such that the ith team in the order beats the $(i+1)$ st team. Find all dominance orders for (a).*

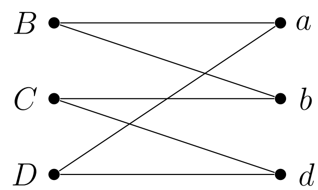
The possible dominance orders are:

(1) Aces - Dogs - Cats - Birds
(2) Birds - Aces - Dogs - Cats
(3) Cats - Birds - Aces - Dogs
(4) Dogs - Cats - Birds - Aces

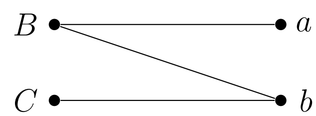
7. *Find all possibles matchings, or explain why none exists for the following graphs.*
(a)



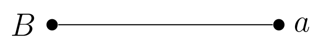
<u>Choose</u>: (A, c)



Choose: (D, d)

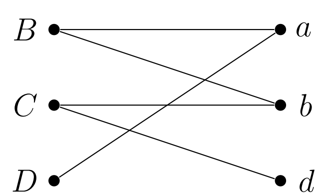


Force: (C, b) (deg(C) = 1)

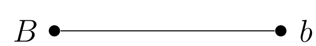


$$\Rightarrow 1) \boxed{\{(A,c),(C,b),(B,a),(D,d)\}}$$

Backtrack from (D,d)

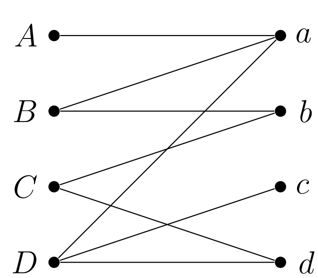


Force: (D,a), (C,d) (deg(D) = deg(d) = 1)

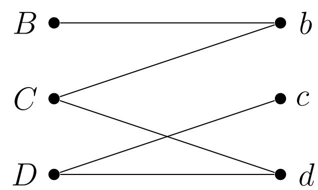


$$\Rightarrow \; 2) \boxed{(D,a),(C,d),(B,b),(A,c)}$$

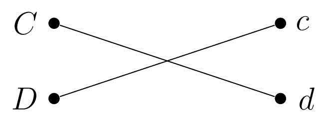
Backtrack from (A, c)



Force: (A,a) (deg(A) = 1)



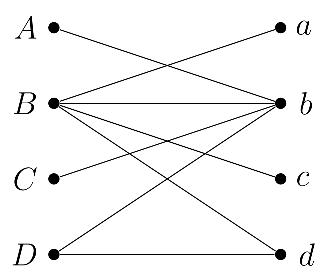
Force: (B,b) (deg(B) = 1)



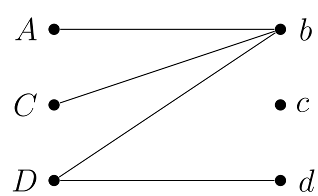
$\Rightarrow$ 3) $\boxed{(A,a),(B,b),(D,c),(C,d)}$

All choices have been exhausted, $\therefore$ all possible solutions have been found.

(b)

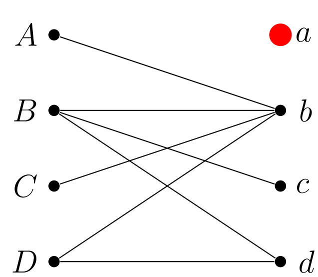


Force: (B,a) ($\deg(a) = 1$)



$\Rightarrow$ No matching exists (either A or C will end up with degree 0 and $\deg(c) = 0$)

Backtrack from (B, a)



$\Rightarrow$ $\boxed{\text{Deleting } (B,a) \text{ results in } \deg(a) = 0 \text{ and } \therefore \text{ no matching exists.}}$

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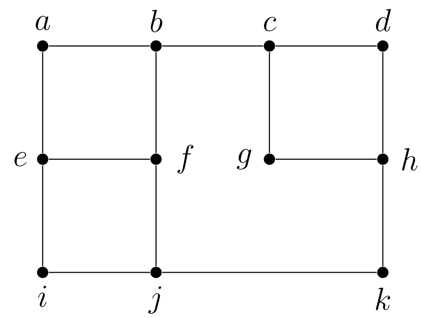
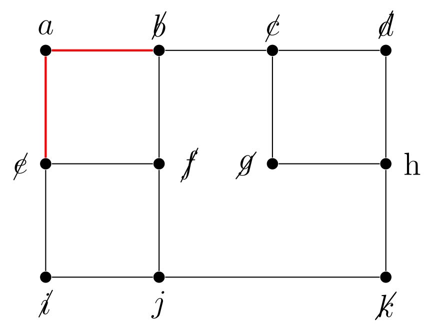
18. *In Figure 1.4, find all sets of three vertices that are adjacent to all the other vertices. Give a careful logical analysis to justify your answer.*



The graph G contains 11 vertices. Let C be the set containing the three vertices. Then, every vertex in G is either in C or adjacent to a vertex in C. Since the maximum possible degree in G is 3, no vertex can cover more than 3 vertices. Also, since the vertices in C account for their self, 8 remaining vertices are left to be covered, implying that potential solutions take the form of

(1) 3 vertices of degree 3 with 1 overlap
or
(2) 2 vertices of degree 3 and 1 vertex of degree 2 with <u>no</u> overlap

<u>Choose</u>: a

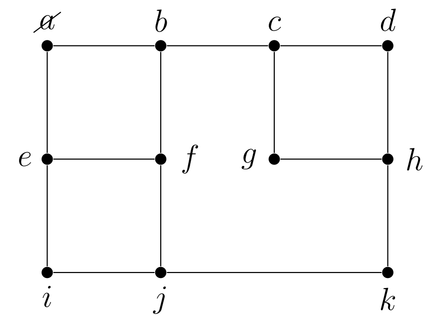


- Eliminates e, c, b, f (By (2): no overlap can happen with a since it's degree 2)
- Eliminates i, g, k, d (By (2): can only have 1 vertex of degree 2)

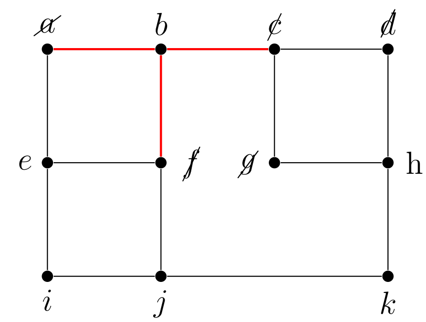
$\Rightarrow$ not possible, c can not be covered

$\Rightarrow$ no solution exists with $|C| = 3$ containing a.

<u>Backtrack</u> from a

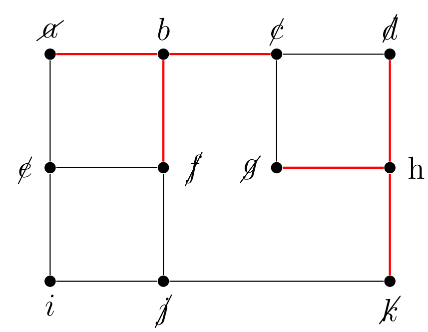


Choose: b

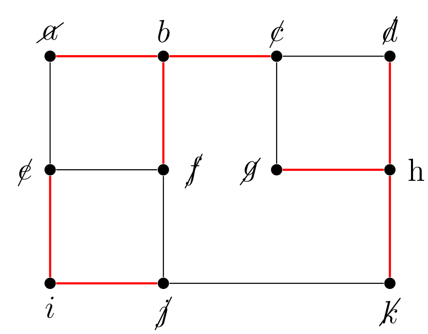


- Eliminates d and g (By (1): d and g are degree 2 but overlap with b)

Choose: h (covers d, g)

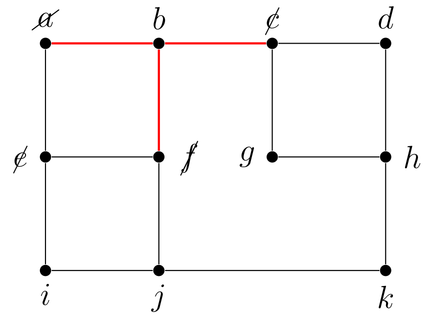


Force: i (covers e, j)



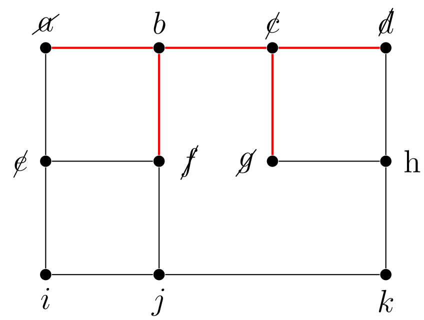
$\Rightarrow$ (1) $\boxed{\{b, h, i\}}$

Backtrack from h



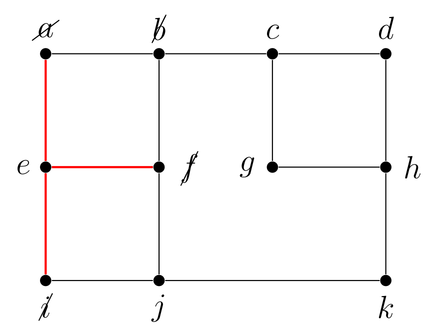
- Eliminates e (By (1), e has 2 overlaps with b)

Force: c



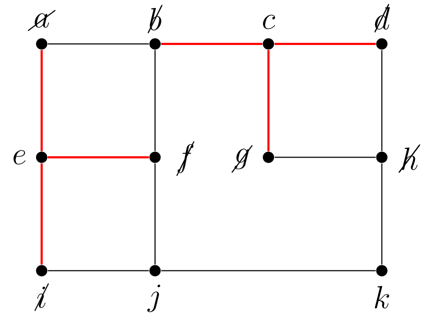
$\Rightarrow$ Not possible by (1), either e, j, or h will cause an extra overlap either with b or c.
$\Rightarrow$ No solution exists with $|C| = 3$ containing b, c

Force: e (covers a)



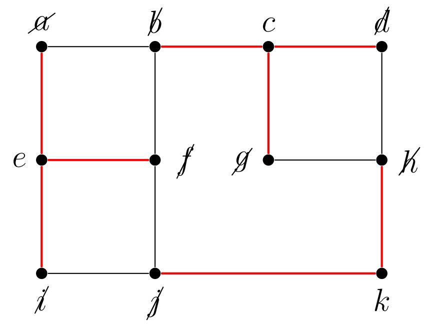
- Eliminates b (by (1), b would cause 2 overlaps with e) - Eliminates i (by (2), i overlaps with e)

Choose: c



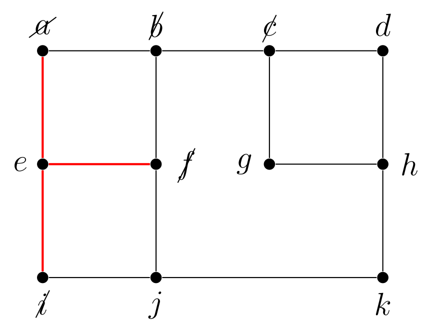
- Eliminates h (by (1), h has 2 overlaps with c)

<u>Force</u>: k (covers j and h)

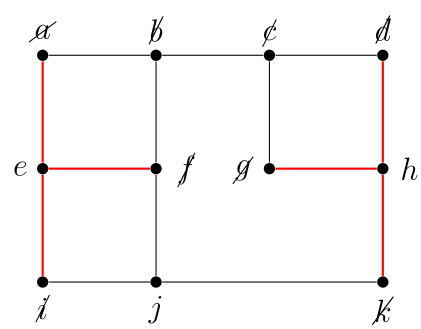


$\Rightarrow$ (2) $\boxed{\{e, c, k\}}$

<u>Backtrack</u> from c



<u>Force</u>: h



$\Rightarrow$ Not possible by (1), j creates 2 overlaps with e.

$\Rightarrow$ No solution exists with $|C| = 3$ containing e, j, and k.

All choices have been exausted and $\therefore$ all solutions have been found.